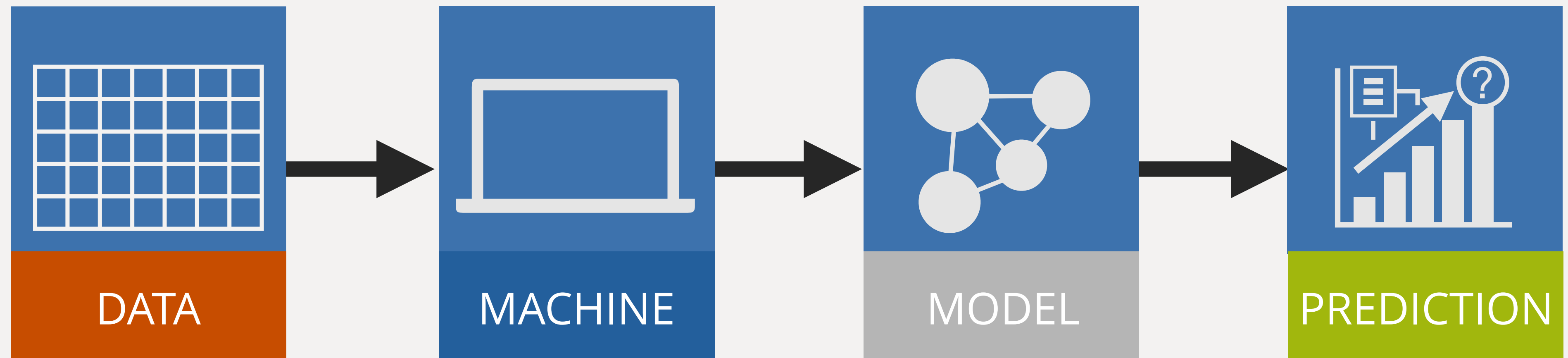




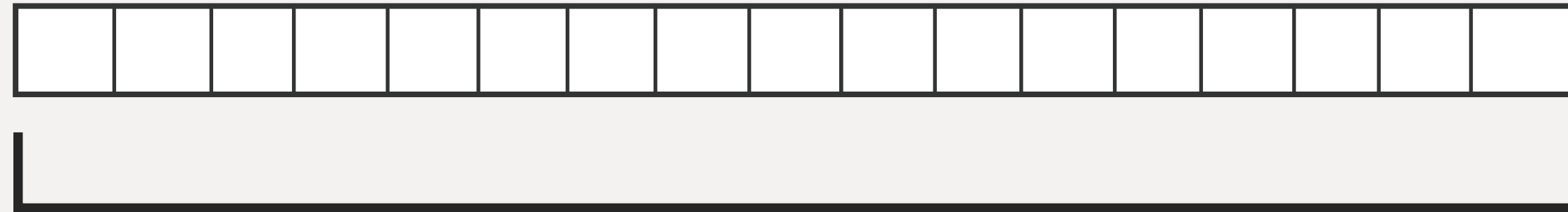
What Is Machine Learning?



In machine learning we want
to teach a machine to learn



Learning a Predictive Model Based on Labeled Data



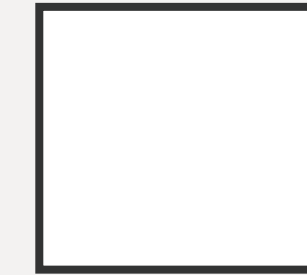
x = data/features for a subject

Learning a Predictive Model Based on Labeled Data

3	12	32	50	7	2	53	32	21	65	43	39	27	6	5	19	12
---	----	----	----	---	---	----	----	----	----	----	----	----	---	---	----	----



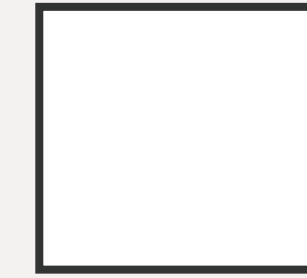
$\mathcal{X}$ = data/features for a subject



y = associated label 0/1

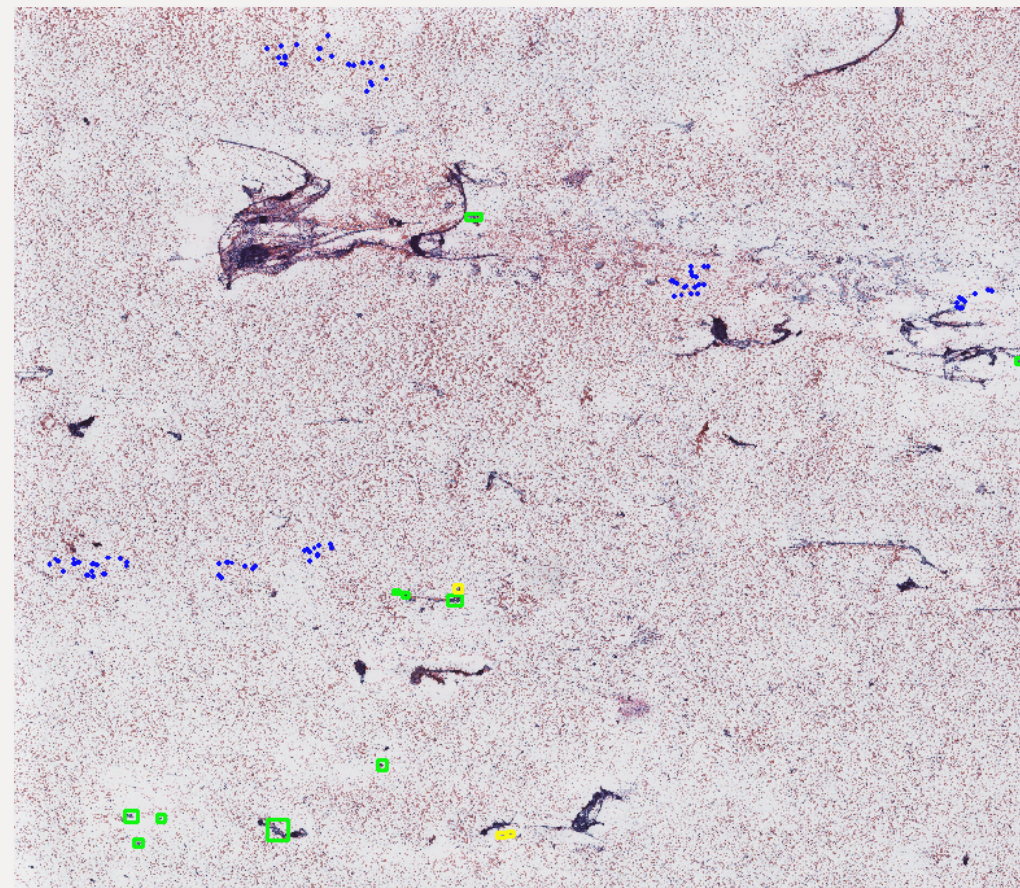
Learning a Predictive Model Based on Labeled Data

3	12	32	50	7	2	53	32	21	65	43	39	27	6	5	19	12

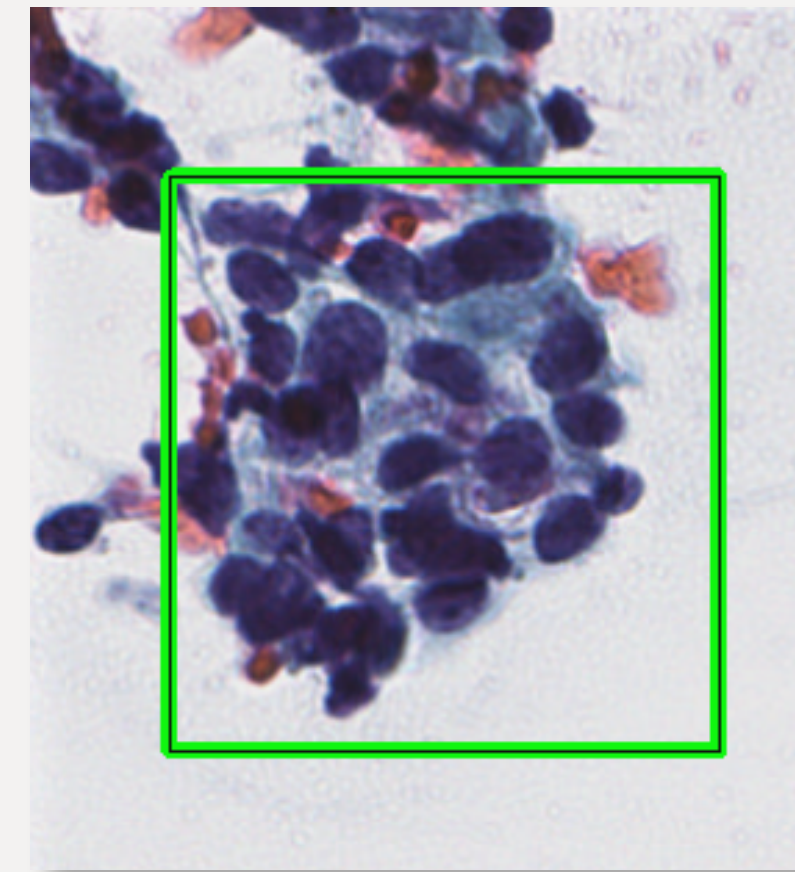


$\mathcal{X}$ = data/features for a subject

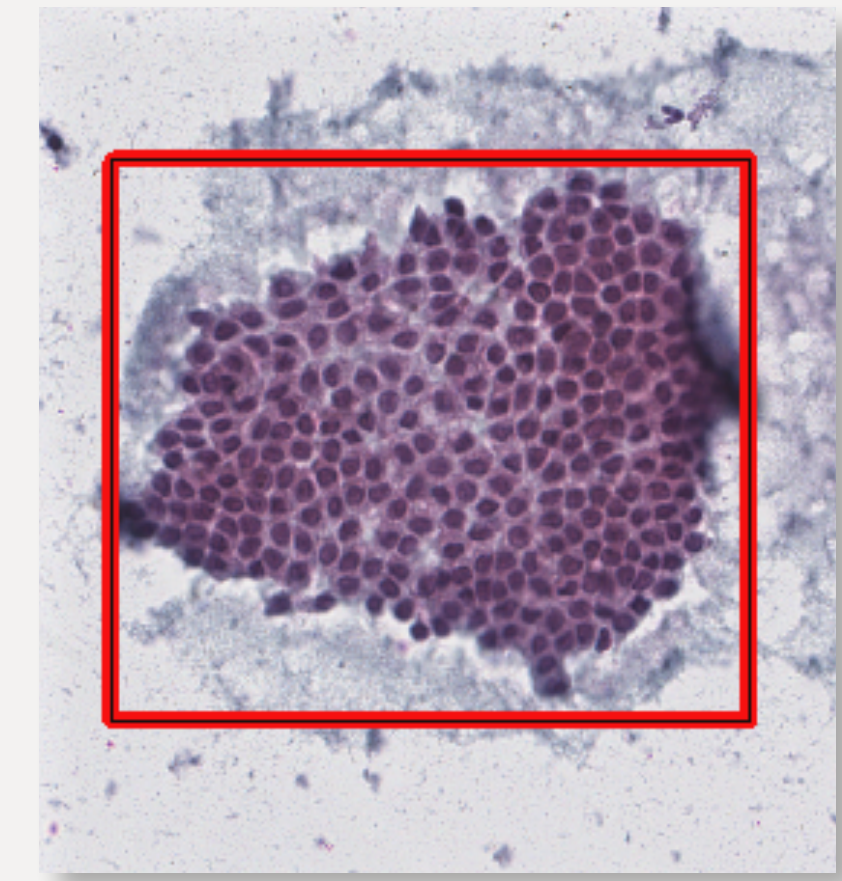
y = associated label 0/1



pixelated image

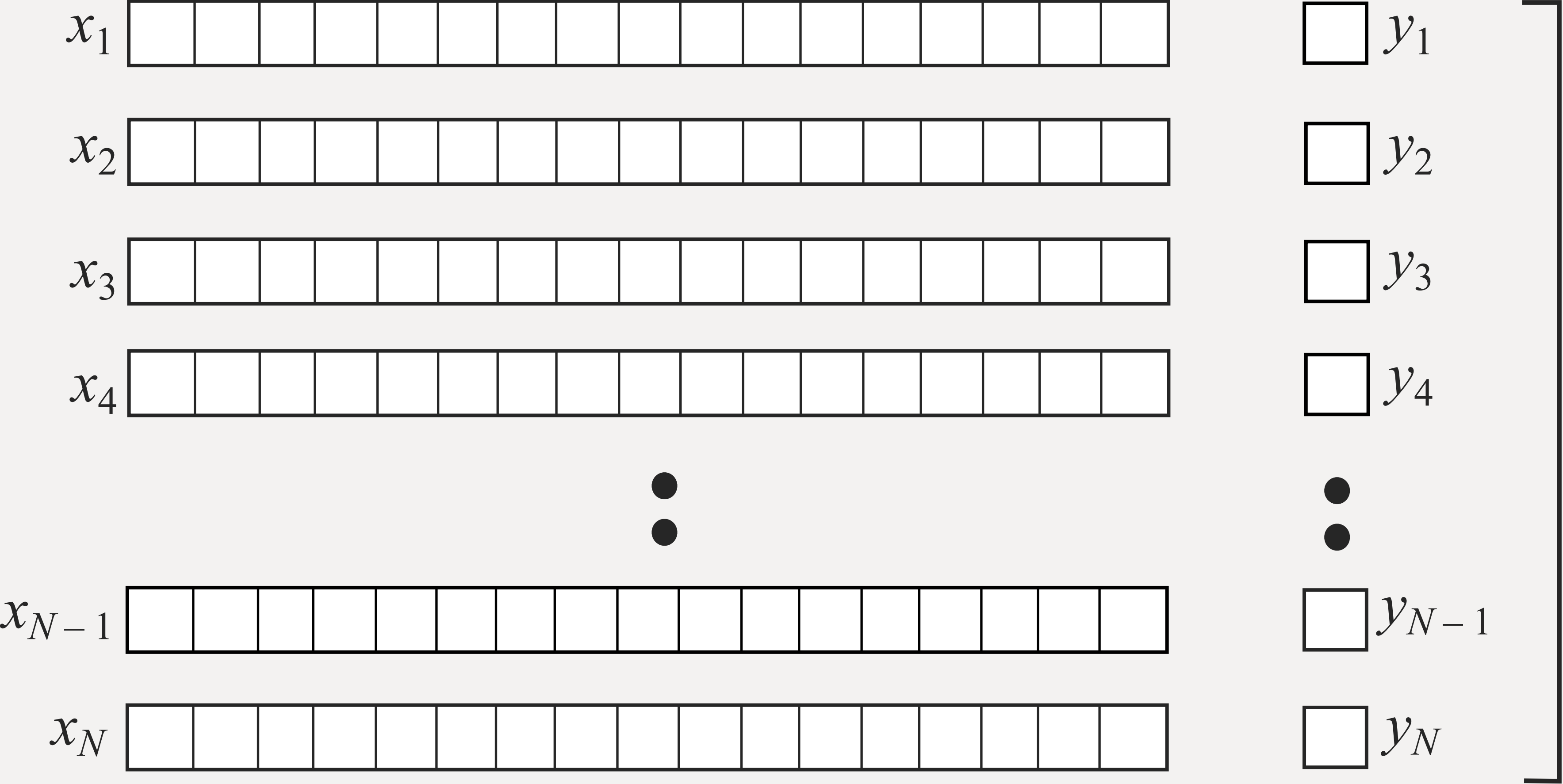


benign



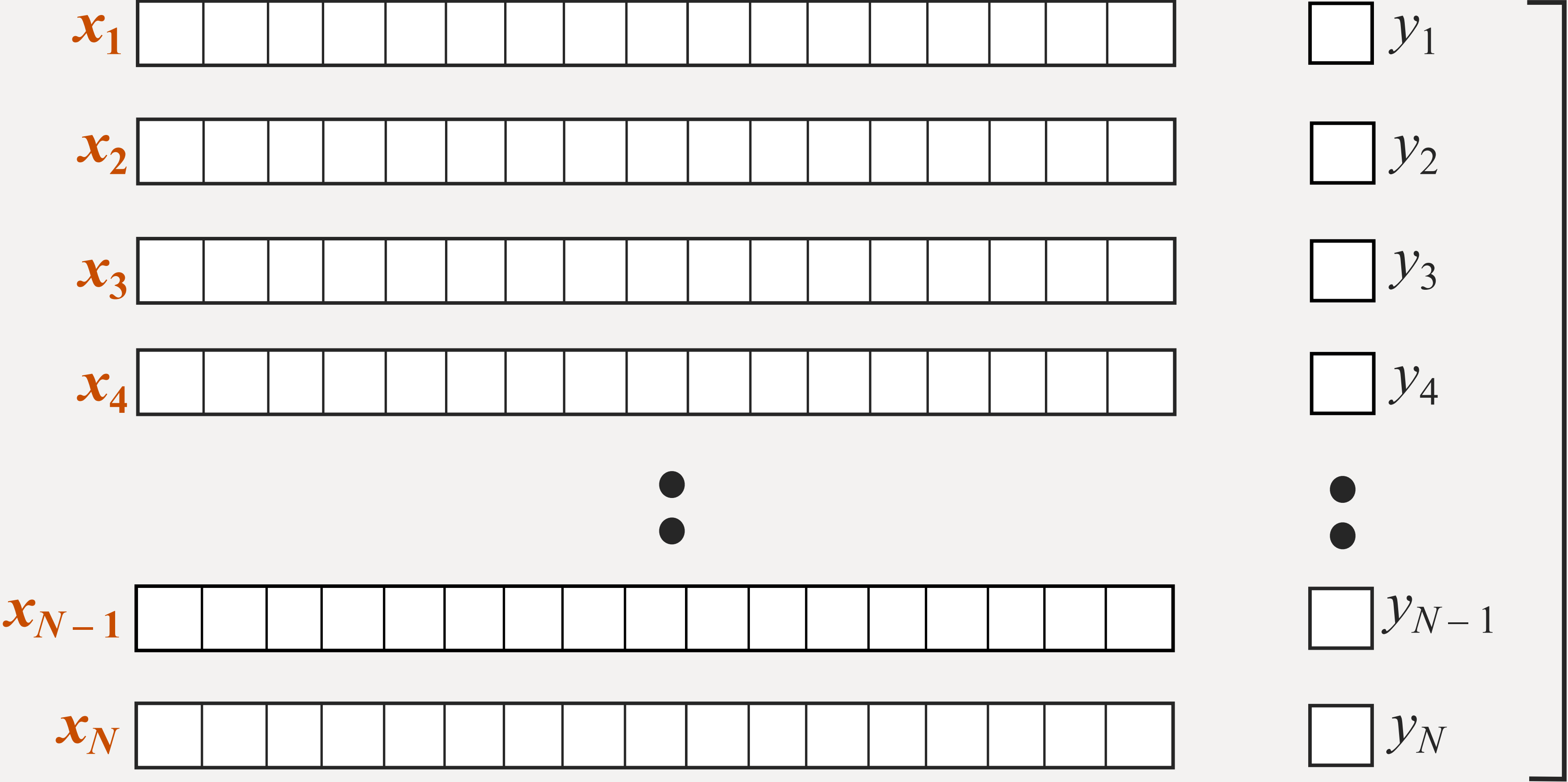
malignant

Training Set



Training Set

$x = \text{data}$



Training Set

 $x = \text{data}$

y = outcome

 x_1 [illegible]

10

 y_1 x_2 [illegible]

11

 y_2 x_3 [illegible]

11

 $\mathcal{Y}_3$ x_4 [illegible]

11

 y_4

• •

•

 x_{N-1} [illegible]

7

$$\mathcal{Y}_{N-1}$$
 x_N [illegible]

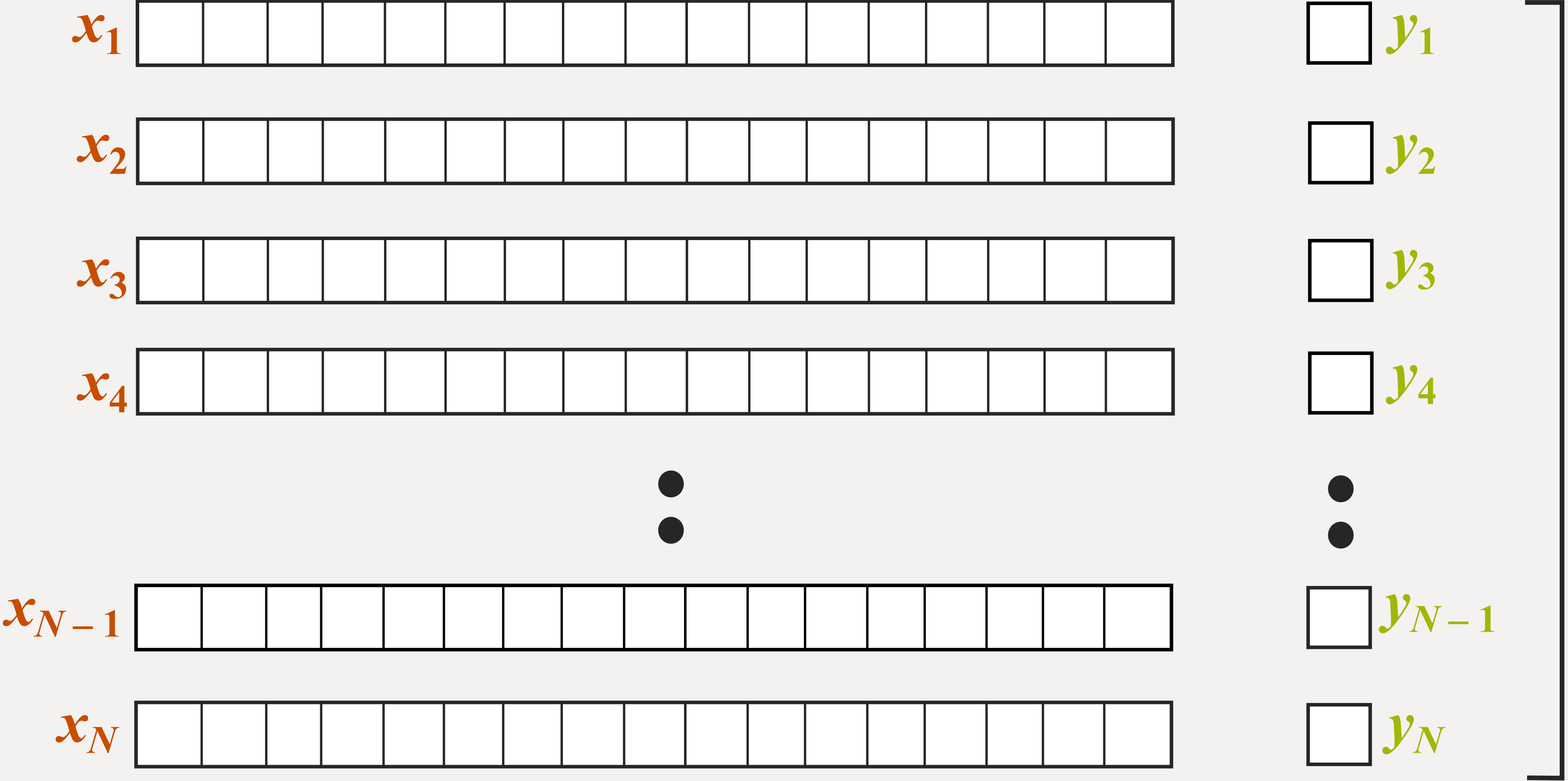
7

 y_N

Training Set

$x = \text{data}$

$y = \text{outcome}$



Will it Rain?

features

outcome

Will it Rain?

x_i = features
for day i



features

Cloud Cover	Humidity	Temperature	Air Pressure

outcome

Will it Rain?

x_i = features
for day i



features

Cloud Cover	Humidity	Temperature	Air Pressure
0.5	80%	75	1.2

outcome

Will it Rain?

x_i = features
for day i



WEATHER

features

Cloud Cover	Humidity	Temperature	Air Pressure
0.5	80%	75	1.2

outcome

Did it Rain
1



RAIN

Will it Rain?

x_i = features
for day i



features

Cloud Cover	Humidity	Temperature	Air Pressure
0.5	80%	75	1.2

outcome

$y_i = 1$, rains

Did it Rain
1



Will it Rain?

x_i = features
for day i



features

Cloud Cover	Humidity	Temperature	Air Pressure
0.5	80%	75	1.2

outcome

Did it Rain
1

$y_i = 1$, rains
 $y_i = 0$, no rain



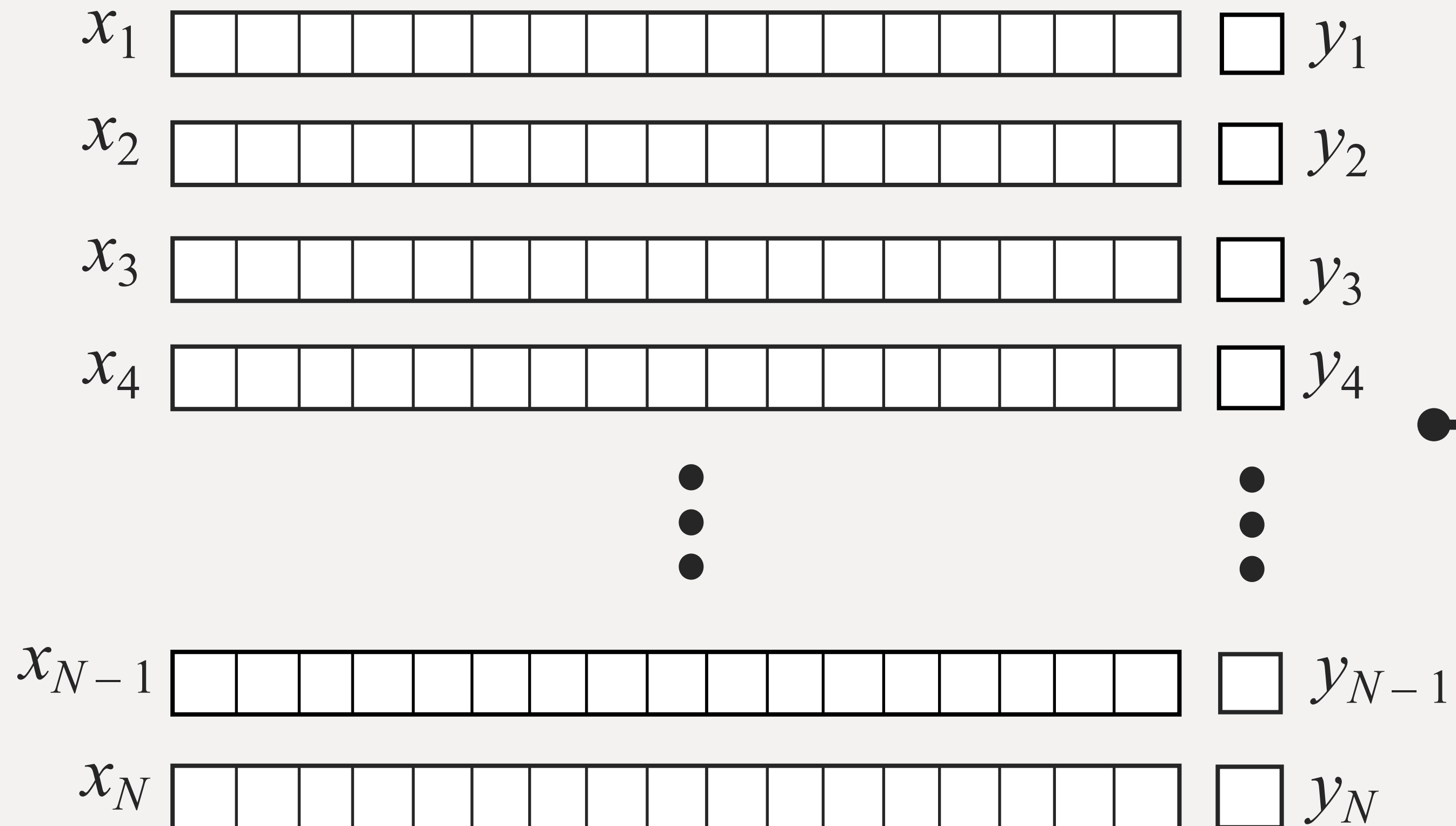
Learned Model Parameters

Training Set

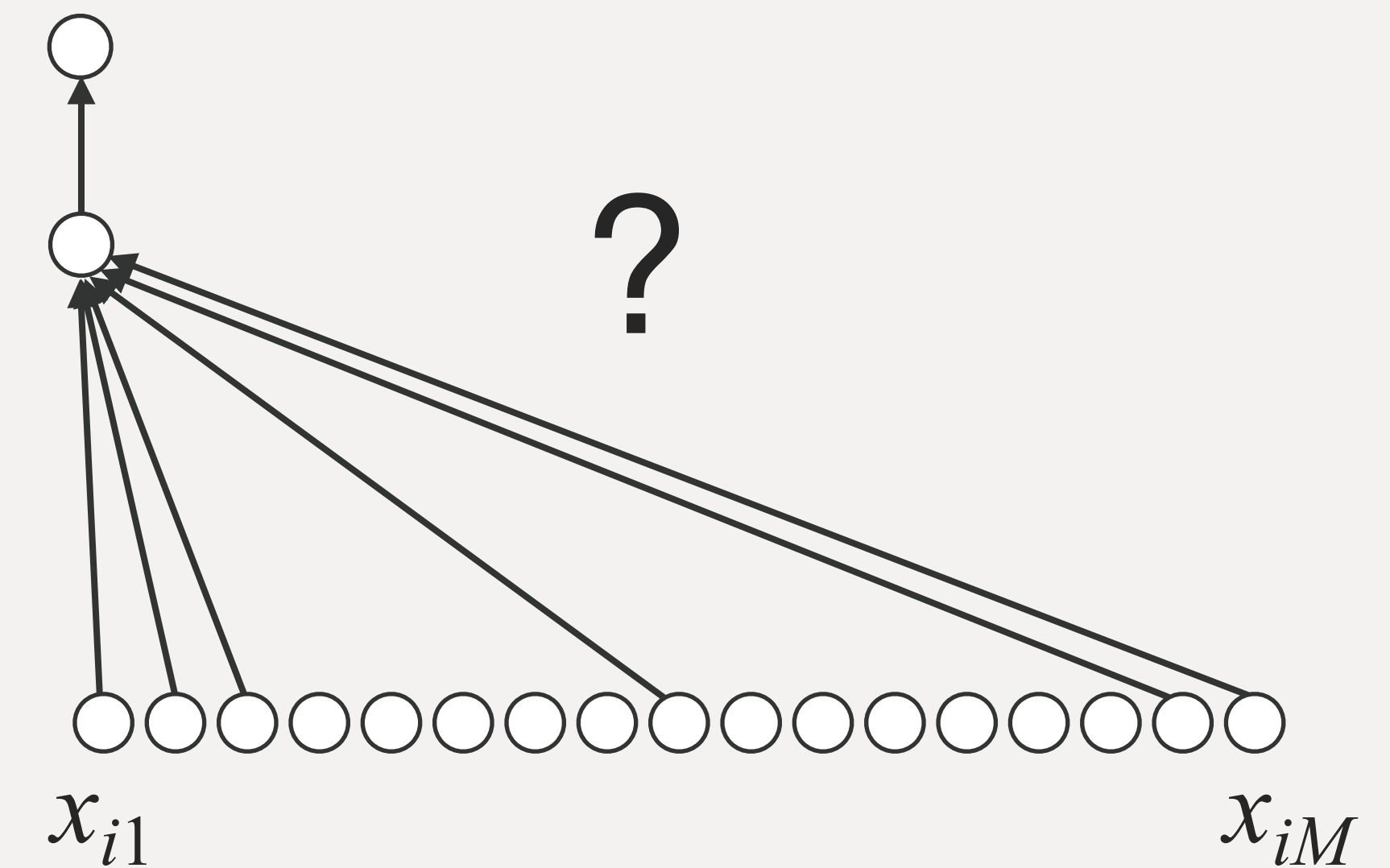
x_1		y_1
x_2		y_2
x_3		y_3
x_4		y_4
x_{N-1}		y_{N-1}
x_N		y_N

Learned Model Parameters

Training Set

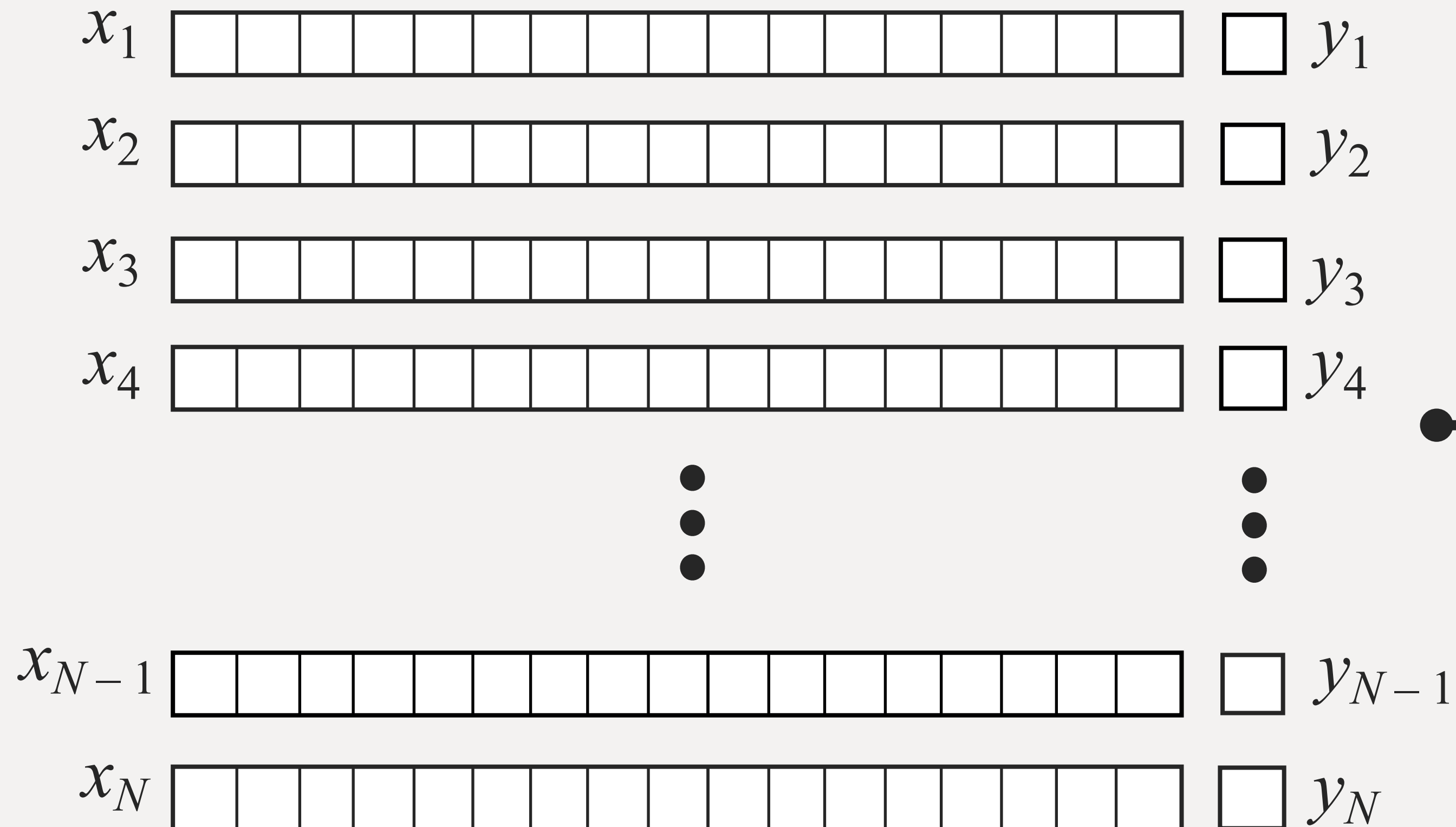


Mathematical Model

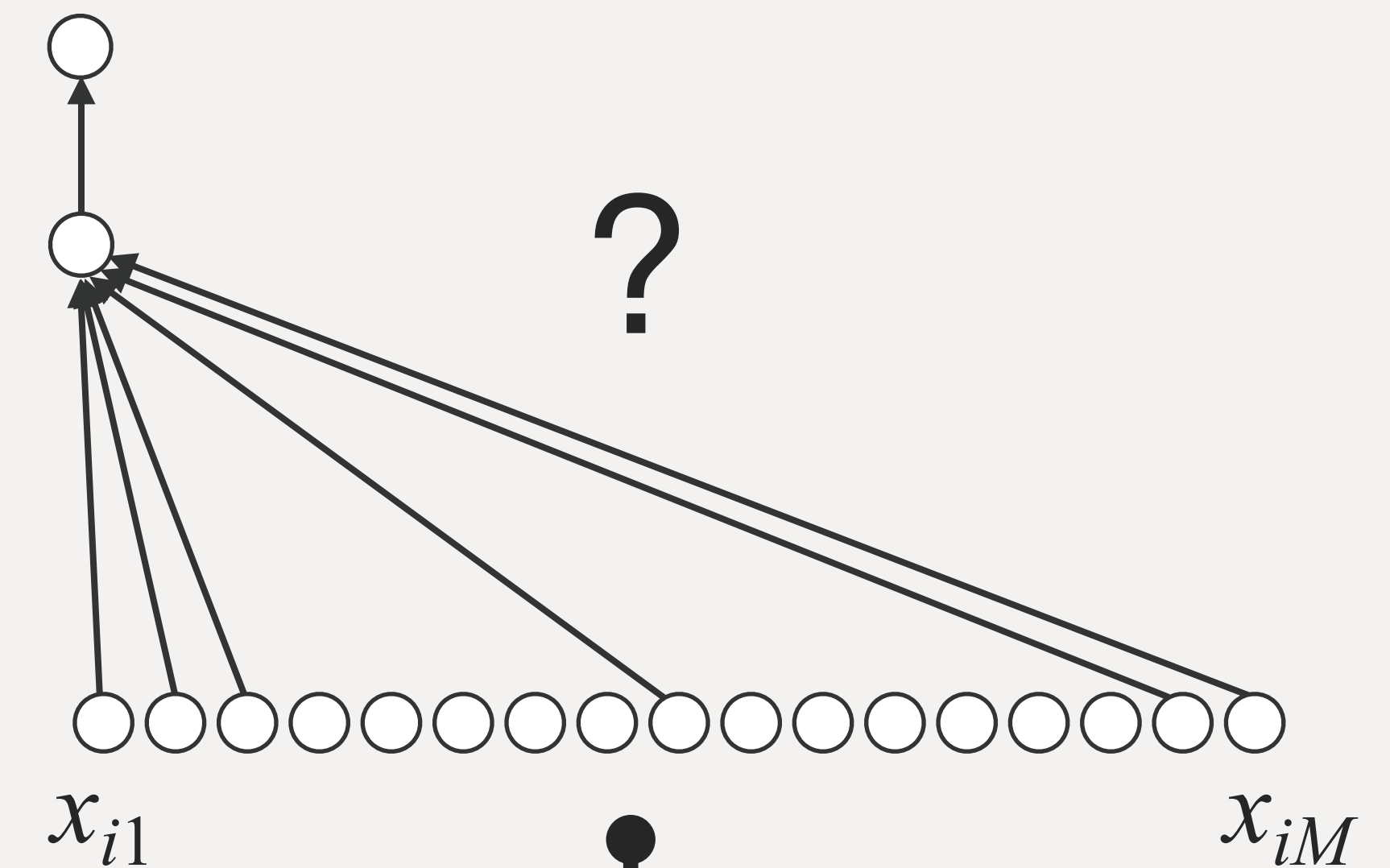


Learned Model Parameters

Training Set



Mathematical Model



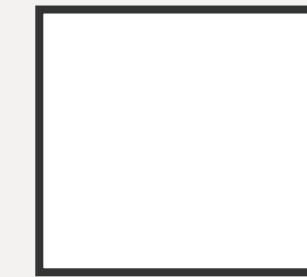
Learned Parameters



Learning a Predictive Model Based on Labeled Data

3	12	32	50	7	2	53	32	21	65	43	39	27	6	5	19	12

$\mathcal{X}$ = data/features for a subject



$\mathcal{Y}$ = associated label 0/1